The two-row $d = 4$ fiber law: reduction to a no-rational-root statement, with a 2-adic tower

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Abstract

For $\lambda = (2m-b, b)$ with $\psi = h_2 + ie_2$, the $d = 4$ fiber-vanishing law asserts $G_\lambda := \langle s_\lambda, \psi^m \rangle = 0 \iff \lambda = (2,2)$. Prior work reduced the two-row case to $\operatorname{Im} G_{(2m-b,b)} \neq 0$ and closed $b \leq 4$. We give a clean reduction $\operatorname{Im}\big((1+su+u^2)^m\big) = u\, H_m(u)$ with $H_m \in \mathbb{Z}[u]$, prove that $I_b(m) := \operatorname{Im} G_{(2m-b,b)}$ is a polynomial of degree b (or $b-1$ if $4 \mid b$) with forced integer roots exactly $\{0, \ldots, \lfloor (b-1)/2 \rfloor\}$, and thereby reduce the law to: the explicit "tail" polynomial Q_b has no integer root $\geq b$. Computation through $b = 24$ shows Q_b is irreducible over $\mathbb{Q}$ (or a half-integer linear factor times an irreducible when $4 \mid b$), so the law follows from the (stronger, cleaner) no-rational-root statement ($\diamond$). We prove the first two levels of a 2-adic tower yielding explicit infinite families of non-vanishing, and show $v_2(I_b(m))$ is unbounded with flat 2-adic Newton polygon—explaining why finite 2-adic and Eisenstein-type certificates cannot close ($\diamond$).

1 Setup and the engine identity

Write $s = 1+i$, $P(u) = (1+su+u^2)^m$, and $G_{(2m-b,b)} = [u^b]\big((1-u)P\big)$. Set $I_b(m) := \operatorname{Im} G_{(2m-b,b)} \in \mathbb{Z}$. Since $s - 1 = i$, with $W := 1 + u + u^2$,

$$1 + su + u^2 = (1 + u + u^2) + iu = W + iu =: A, \qquad B := \bar{A} = W - iu,$$

so that $A + B = 2W$ and $AB = W^2 + u^2$.

2 The clean reduction

Theorem 1. *$\operatorname{Im}(A^m) = u\, H_m(u)$, where $H_m(u) = \dfrac{A^m - B^m}{A - B} = \sum_{j=0}^{m-1} A^j B^{m-1-j} \in \mathbb{Z}[u]$ is symmetric in A, B, hence a polynomial in $2W$ and $W^2 + u^2$. Consequently*

$$I_b(m) = [u^{b-1}]\big((1-u)\, H_m(u)\big).$$

Proof. As $B = \bar{A}$, $A^m - B^m = 2i \operatorname{Im}(A^m)$ and $A - B = 2iu$, so $H_m = \operatorname{Im}(A^m)/u$. Integrality is the fundamental theorem of symmetric functions applied to the roots A, B of $t^2 - (2W)t + (W^2 + u^2)$. Finally $I_b(m) = [u^b]\big((1-u) \operatorname{Im} P\big) = [u^b]\big((1-u)\, u\, H_m\big) = [u^{b-1}]\big((1-u) H_m\big)$. $\square$

The recurrence $h_n = 2W\, h_{n-1} - (W^2 + u^2) h_{n-2}$ ($h_0 = 1, h_1 = 2W$, $H_m = h_{m-1}$) gives $H_m = (W^2 + u^2)^{(m-1)/2}\, U_{m-1}\big(W/\sqrt{W^2 + u^2}\big)$ with U the Chebyshev polynomial of the second kind.

3 Degree and the forced integer roots

Theorem 2. (a) $\deg_u(\operatorname{Im} P) = 2m-1$ *with leading coefficient* m*; hence* $\deg_u\big((1-u)\operatorname{Im} P\big) = 2m$.
(b) $I_b(m) = 0$ *for every integer* m *with* $2m < b$*, i.e.* $m \in \{0, 1, \ldots, \lfloor (b-1)/2 \rfloor\}$.
(c) *As a polynomial in* m*,* $\deg_m I_b = b$ *if* $b \not\equiv 0 \pmod 4$ *and* $b-1$ *if* $4 \mid b$*; in particular* I_b *is a nonzero polynomial.*

Proof. (a) $\operatorname{Im} P = \sum_{k \text{ odd}} (-1)^{(k-1)/2} \binom{m}{k} u^k W^{m-k}$; the k-th term has degree $2m-k$, maximal at $k=1$ with leading coefficient $\binom{m}{1} = m$. (b) $(1-u)\operatorname{Im} P$ has degree $2m$, so $[u^b] = 0$ once $b > 2m$. (c) For fixed l, $[u^l]P = \sum_k \binom{m}{k} i^k T(m-k, l-k)$ with $T(N,j) = [u^j]W^N$. Since $\binom{m}{k} \sim m^k/k!$ and $T(m-k, l-k) \sim m^{l-k}/(l-k)!$, the k-th summand is $\sim \frac{i^k}{k!(l-k)!} m^l$, whence $[u^l]P \sim \frac{(1+i)^l}{l!} m^l$ and $\operatorname{Im}[u^l]P \sim \frac{2^{l/2}\sin(\pi l/4)}{l!} m^l$. This m^l-coefficient vanishes iff $4 \mid l$; applying it to $I_b = \operatorname{Im}[u^b]P - \operatorname{Im}[u^{b-1}]P$ gives the stated degree. $\square$

Thus $I_b(m) = \Big(\prod_{r=0}^{\lfloor (b-1)/2 \rfloor} (m-r)\Big) Q_b(m)$ with $\deg Q_b = \lfloor b/2 \rfloor$ (resp. $\lfloor b/2 \rfloor - 1$ if $4 \mid b$). For $m \geq b$ the product is nonzero, so the two-row law is equivalent to

$$(\star) \qquad Q_b(m) \neq 0 \text{ for all integers } m \geq b.$$

4 Reduction to "no rational root"

Theorem 3 (verified $b \leq 24$)**.** *If* $b \not\equiv 0 \pmod 4$*,* Q_b *is irreducible over* $\mathbb{Q}$*. If* $4 \mid b$*,* $Q_b = (2m - (2b-1))\, R_b(m)$ *with* R_b *irreducible of degree* $\lfloor b/2 \rfloor - 1$*; the linear factor gives the simple rational root* $m = (2b-1)/2$*, a half-integer. In all cases* Q_b *has no integer root, so the two-row law holds for* $b \leq 24$*.*

The factor $2m - (2b-1)$ for $4 \mid b$ reflects $I_b\big((2b-1)/2\big) = 0$, verified directly for $b = 4, 8, \ldots, 28$. Hence the entire remaining gap is the single assertion

$$(\diamond) \qquad \text{for } b \geq 5,\ Q_b \text{ has no rational root, except the simple } (2b-1)/2 \text{ when } 4 \mid b.$$

Why cheap certificates fail. For the primitive integer model of Q_b, both the constant and leading coefficients are odd: the 2-adic Newton polygon is the flat slope-0 segment, so all roots are 2-adic units (no 2-adic ramification, no Eisenstein prime at 2). And since $\deg Q_b \to \infty$, no fixed prime p can give "no root mod p" uniformly. So $(\diamond)$ requires genuine arithmetic input.

5 The 2-adic tower

Mod 2, $W^2 + u^2 \equiv 1 + u^4$.

Theorem 4. *Using* $H_m = \sum_c (-1)^c \binom{m-1-c}{c} (2W)^{m-1-2c} (W^2+u^2)^c$*:*

Level 1. *If* m *is even, all coefficients of* H_m *are even, so* $I_b(m)$ *is even. If* m *is odd,* $H_m \equiv (1+u^4)^{(m-1)/2} \pmod 2$*, whence with* $M = (m-1)/2$*,*

$$I_b(m) \equiv \begin{cases} \binom{M}{(b-1)/4} & b \equiv 1 \\ \binom{M}{(b-2)/4} & b \equiv 2 \\ 0 & b \equiv 0, 3 \end{cases} \pmod 2 \quad (\text{mod } 4 \text{ on } b).$$

Level 2. *If* $m \equiv 0 \pmod 2$*,* $M' = (m-2)/2$*, then* $H_m \equiv 2(-1)^{M'}(m/2)\, W\, (1+u^4)^{M'} \pmod 4$*, and since* $(1-u)W = 1-u^3$*,*

$$\tfrac{1}{2} I_b(m) \equiv \begin{cases} (m/2)\binom{M'}{(b-1)/4} & b \equiv 1 \\ (m/2)\binom{M'}{(b-4)/4} & b \equiv 0 \\ 0 & b \equiv 2,3 \end{cases} \pmod 2.$$

Corollary 5. *$I_b(m) \neq 0$ whenever (m odd, $b \equiv 1, 2$, $\binom{(m-1)/2}{\lfloor (b-1)/4 \rfloor}$ odd) or ($m \equiv 2 \pmod 4$, $b \equiv 0, 1$, level-2 binomial odd). By Lucas these are infinite families for each admissible b.*

Remark 6. $v_2(I_b(m))$ is unbounded (it reaches 11 at $(m, b) = (18, 6), (18, 10)$), so no finite truncation of the tower closes $(\diamond)$: m can be 2-adically arbitrarily close to a (2-adic unit) root of Q_b, forcing arbitrarily large v_2—yet never equal, as those roots are irrational. This is the 2-adic shadow of $(\diamond)$.

6 Status

Theorems 1, 2, 4 are rigorous; Theorem 3 is verified through $b = 24$. The law $(\star)$ follows from $(\diamond)$, an irreducibility/no-rational-root statement about an explicit integer-valued polynomial family of degree $\lfloor b/2 \rfloor$ with flat 2-adic Newton polygon—likely a disguised classical orthogonal family, whose identification is the natural next step.